

```
const SHA256 = require("sha256");

class Block {
  constructor(index, timestamp, data, prevHash = "") {
    this.index = index;
    this.timestamp = timestamp;
    this.data = data;
    this.prevHash = prevHash;
    this.hash = this.calculateHash();
  }

  calculateHash() {
    return SHA256(
      this.index +
      this.prevHash +
      this.timestamp +
      JSON.stringify(this.data)
    ).toString();
  }
}

class Blockchain {
  constructor() {
    this.chain = [this.createGenesisBlock()];
  }

  createGenesisBlock() {
    return new Block(0, "15/07/1974", "Genesis Block", 0);
  }

  getLatestBlock() {
    return this.chain[this.chain.length - 1];
  }

  addBlock(newBlock) {
    newBlock.prevHash = this.getLatestBlock().hash;
    newBlock.hash = newBlock.calculateHash();
    this.chain.push(newBlock);
  }
}

let firbitCoin = new Blockchain();
firbitCoin.addBlock(new Block(1, "10/02/15", { amount: 400 }));
firbitCoin.addBlock(new Block(2, "20/02/15", { amount: 800 }));

console.log(JSON.stringify(firbitCoin, null, 4));
```

npm init -y:

```
PS C:\Users\SHRUTIKA MHETAR> npm init -y
Wrote to C:\Users\SHRUTIKA MHETAR\package.json:

{
  "name": "shrutika-mhetar",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

npm install sha256:

```
PS C:\Users\SHRUTIKA MHETAR> npm install sha256
added 3 packages, and audited 4 packages in 6s
found 0 vulnerabilities
```

node bct.js:

```
Node.js v24.12.0
PS C:\Users\SHRUTIKA MHETAR> node bct.js
{
  "chain": [
    {
      "index": 0,
      "timestamp": "15/07/1974",
      "data": "Genesis Block",
      "prevHash": 0,
      "hash": "65cbc39552d554dc7e432fc8c1dd14a4f7e60e50901b81d3e90fab9b2fddbdfff"
    },
    {
      "index": 1,
      "timestamp": "10/02/15",
      "data": {
        "amount": 400
      },
      "prevHash": "65cbc39552d554dc7e432fc8c1dd14a4f7e60e50901b81d3e90fab9b2fddbdfff",
      "hash": "1fb975e318f7aa3687823f388a57b062ec8b2d9296ae9dc796d1e1e457f6898e"
    },
    {
      "index": 2,
      "timestamp": "20/02/15",
      "data": {
        "amount": 800
      },
      "prevHash": "1fb975e318f7aa3687823f388a57b062ec8b2d9296ae9dc796d1e1e457f6898e",
      "hash": "e73d392ea5b70f2929ac068b55b4d897213090dc9972e1e8fdec13dbe5668e6b"
    }
  ]
}
```